AQA 3.2.2 and 3.2.3 Tests for ions (required practical 4)

AQA A-level Chemistry · Revision questions · Sheet 5 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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18 marks · about 25 minutes

Key information

- Halides with HNO_3-acidified $AgNO_3(aq)$: Cl^- white, Br^- cream, I^- yellow; F^- no precipitate
- Group 2 cations: NaOH(aq) gives white $Mg(OH)_2$; $H_2SO_4(aq)$ gives white $BaSO_4$ with Ba^{2+}
- NH_4^+: warm with NaOH(aq), damp red litmus turns blue; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
- CO_3^{2-}: dilute acid gives effervescence; CO_2 turns limewater cloudy
- SO_4^{2-}: add HCl-acidified $BaCl_2(aq)$ — white precipitate; the acid removes CO_3^{2-} first

Section A: Short-answer questions

01 Describe a test for carbonate ions in a solid. Give the observations for a positive result.

[2 marks]

02 A white solid dissolves in water. The solution gives a white precipitate with acidified barium chloride, and when warmed with sodium hydroxide solution it releases a gas that turns damp red litmus paper blue. Identify the solid.

[2 marks]

03 In the test for sulfate ions, dilute hydrochloric acid is added before barium chloride solution. Explain why.

[1 mark]

04 In the test for a halide ion in aqueous solution, dilute nitric acid is added before the addition of silver nitrate solution. Why is nitric acid added?

[1 mark]

A It increases the concentration of nitrate ions.
B It prevents the precipitation of silver compounds other than halides.
C It prevents the silver nitrate being precipitated.
D It provides the acidic solution required for precipitation.

05 An aqueous solution of a salt gives a white precipitate when mixed with aqueous silver nitrate and when mixed with dilute sulfuric acid. Which could be the formula of the salt?

[1 mark]

A $BaCl_2$
B $(NH_4)_2SO_4$
C KCl
D $Sr(NO_3)_2$

06 Which row shows the colours of the precipitates formed when acidified silver nitrate is added to solutions of chloride, bromide and iodide ions? (chloride / bromide / iodide)

[1 mark]

A white / yellow / cream
B cream / white / yellow
C white / cream / yellow
D yellow / cream / white

Section B: Longer questions

07 A student completes some test-tube reactions on five solutions, P, Q, R, S and T. Each test is done on separate samples of each solution. Observations are shown in the table.

Solution	Test 1 Add a few drops of $H_2SO_4(aq)$	Test 2 Add $HNO_3(aq)$ then a few drops of $AgNO_3(aq)$	Test 3 Add a few drops of NaOH(aq)	Test 4 Add a few drops of $Cl_2(aq)$
P	White precipitate	Cream precipitate	No visible change	Orange solution
Q	Effervescence	Effervescence and white precipitate	No visible change	No visible change
R	No visible change	No visible change	White precipitate	No visible change
S	White precipitate		No visible change	Dark brown solution
T	No visible change	White precipitate	White precipitate	No visible change

07.1 Which solution contains carbonate ions?

[1 mark]

07.2 Which of $MgBr_2$, $BaBr_2$, $MgCl_2$ or $BaCl_2$ could be the compound in solution P?

[1 mark]

07.3 What observation is expected in Test 2 for solution S?

[1 mark]

07.4 Which of $MgCl_2$, $BaCl_2$, $MgBr_2$ or $BaBr_2$ could be the compound in solution T?

[1 mark]

08 The following pairs of compounds, each in aqueous solution, can be distinguished by simple test-tube reactions.

Give a reagent, or combination of reagents, that can be added to the solutions in each pair to distinguish between them in a single reaction. State what is observed in each case.

08.1 NaCl(aq) and $BaCl_2(aq)$

[3 marks]

08.2 NaCl(aq) and $Na_2CO_3(aq)$

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	Add dilute acid (e.g. HCl) — effervescence gas turns limewater cloudy / milky	*Source: Practice question (written for this set)*	2
02	contains SO_4^{2-} **and** NH_4^+ $(NH_4)_2SO_4$ / ammonium sulfate	*Source: Practice question (written for this set)*	2
03	To remove carbonate ions, which would also give a white precipitate ($BaCO_3$)	*Source: Practice question (written for this set)*	1
04	**B**	*Source: 7405/3 Nov 2020 Q15*	1
05	**A**	*Source: 7405/3 Jun 2019 Q16*	1
06	**C**	*Source: Practice question (written for this set)*	1
07.1	Solution Q	*Source: 7404/1 Jun 2023 Q19–21, Q23* Original items are multiple choice (P, Q, R, S)	1
07.2	$BaBr_2$		1
07.3	Pale yellow precipitate		1
07.4	$MgCl_2$		1
08.1	Reagent: H_2SO_4 / Na_2SO_4 / any soluble sulfate NaCl: no (visible) change $BaCl_2$: white ppt / white solid	*Source: 7404/1 Jun 2019 Q08* If reagent incorrect cannot score observations Do not allow 'nothing' or 'no observation'	3
08.2	Reagent: H_2SO_4 / HCl / HNO_3 NaCl: no (visible) change Na_2CO_3: effervescence / bubbles / fizzing	Or acidified $AgNO_3$: white ppt with NaCl, effervescence with Na_2CO_3 Allow other reagents (e.g. $BaCl_2$) that distinguish in a single reaction	3